

Extending GRB redshift estimation to optical-plateau bursts: methodology and results

P. Lin and collaborators

(affiliations)

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ABSTRACT

Aims.
Methods.
Results.
Conclusions.

1. Methodology

We build directly on the machine-learning (ML) redshift estimator of Narendra et al. (2025), which trained a Super-Learner ensemble on the X-ray-plateau parameters of long gamma-ray bursts (LGRBs). The central extension presented here is the incorporation of *optical-plateau* LGRBs into the training sample: a sizeable population of bursts possess a well-measured optical afterglow plateau but no X-ray plateau, and folding them onto the common X-ray parameter scale enlarges the labelled sample. Because the optical- and X-ray-domain plateau parameters are physically distinct but tightly correlated, this requires an explicit cross-calibration step before the paper’s downstream pipeline (feature selection, formula generation, robust outlier removal, ensemble learning, and Monte Carlo error propagation) can be applied. Each stage is described below; unless stated otherwise the response variable is $\log_{10}(z+1)$ and all logarithms are base 10, following Narendra et al. (2025).

1.1. Data samples

We use two catalogues of LGRBs with spectroscopic redshift. The X-ray sample contains 223 bursts whose afterglow plateau was fitted in the 0.3–10 keV Swift XRT band with the Willingale et al. (2007) (W07) function, from which we take the four plateau parameters $\log F_a$ (plateau end flux), $\log T_a$ (plateau end time), α (post-plateau temporal decay index) and β (plateau spectral index), together with the prompt-emission quantities $\log T_{90}$, $\log \text{Fluence}$, $\log \text{Peak-Flux}$, the prompt photon index, the X-ray spectral index Γ , and the hydrogen column density $\log N_H$. The optical sample contains 161 LGRBs whose plateau was instead measured in the optical band, providing optical-domain counterparts of the same four plateau parameters. Of these, **78** bursts (“anchors”) appear in both catalogues and constrain the optical→X-ray calibration, while **83** are optical-only and are the targets of the projection. The ten candidate features listed above are identical to those of Narendra et al. (2025); see their Sect. 3 and Fig. 1 for the plateau definition.

1.2. Cross-calibration and catalogue combination

For each of the four shared plateau parameters we fit a linear optical→X-ray calibration on the 78 anchors,

$$y = mx + b, \quad (1)$$

where x and y are the optical- and X-ray-domain values. We treat the reported 1σ measurement uncertainties on both axes together with an intrinsic scatter s through a Gaussian likelihood with total variance

$$V = \sigma_y^2 + m^2 \sigma_x^2 + s^2, \quad (2)$$

sampling the parameters $\theta = (m, b, \ln s)$ with flat priors using the affine-invariant ensemble sampler **emcee**. We run 32 walkers for 3000 steps, discarding the first 500 as burn-in and thinning by 15, and pool **four** independently seeded chains per parameter into a single posterior; the point estimate is the posterior median and the 1σ error the posterior standard deviation. The fit is fully deterministic under fixed seeds. The optical-only bursts are then projected onto the X-ray scale by full posterior-predictive propagation: for every posterior draw (m_i, b_i, s_i) each optical measurement is perturbed within its 1σ error, mapped through Eq. (1) with an added scatter term, and summarised by the median (point estimate) and half the 16th–84th percentile range (1σ). When a burst has both an X-ray and a projected optical value for a given column, the X-ray measurement is preferred and the projection serves only as a fallback. The combined catalogue at this stage contains **306** bursts (223 X-ray plus 83 projected optical-only). Figure 1 shows the calibration.

1.3. Quality and non-negotiable outlier cuts

Two cuts precede feature selection. First, at the combination stage we remove any optical-only burst whose relative error $|\sigma|/|\text{value}|$ exceeds 0.5 on any of the four calibrated parameters, on either the original-optical or the projected-X-ray side; this reduces the sample to **296**. Second, we apply a set of non-negotiable physical cuts. We restrict to long bursts via $\log T_{90} > \log 2$ (285 bursts remain) and then remove bursts with unphysical parameter values

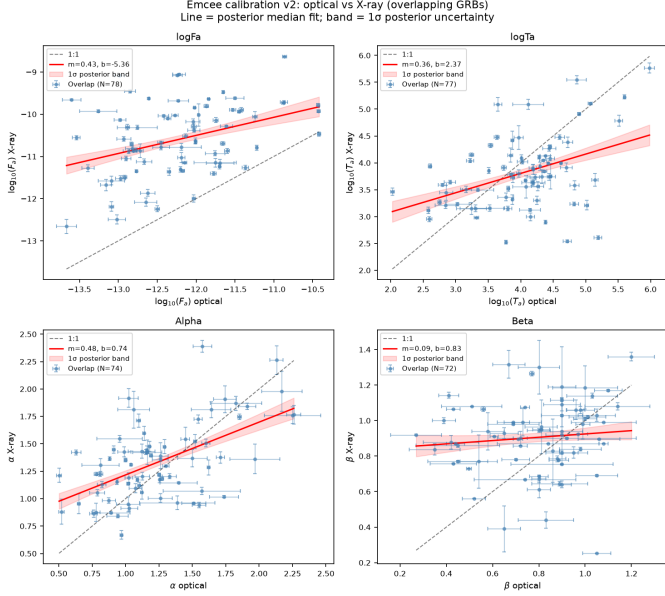


Fig. 1. Optical versus X-ray plateau parameters ($\log F_a$, $\log T_a$, α , β) on the 78 anchor GRBs, with the `emcee` optical→X-ray calibration (Eq. 1) overplotted.

($\alpha > 3$, $\beta > 2$, $\log T_{90} > 6$, or a negative photon index), together with a relative-error cut $|\sigma|/|\text{value}| > 0.5$ that is deliberately restricted to the four *fitted* plateau parameters (α , β , $\log F_a$, $\log T_a$). We do not apply this error cut to the directly measured catalogue quantities (T_{90} , photon index, fluence, N_H , peak flux): a large relative error there reflects genuine measurement noise rather than an unreliable calibration fit, and cutting on it would preferentially discard faint, high-redshift bursts. This stage flags four bursts — GRB 090516A, GRB 100621A, GRB 131117A, and GRB 230328B — leaving **281** bursts for training. The same counts and the same four flagged bursts are obtained independently for the alternative optimal-transport combination (not discussed here).

1.4. Feature selection

Following Narendra et al. (2025, Sect. 4.2.1) we rank the ten candidate features with the least absolute shrinkage and selection operator (LASSO; Tibshirani 1996). We run `cv.glmnet` ($\alpha = 1$) 100 times, each time recording the coefficient vector at λ_{1se} , average the 100 vectors, and order the features by mean absolute coefficient. The seven highest-ranked features are retained: $\log F_a$, $\log \text{PeakFlux}$, $\log N_H$, photon index, $\log T_a$, α , and $\log T_{90}$ (Fig. 2), matching the seven-feature choice of the parent paper.

1.5. Formula generation and selection

From the seven selected features and their seven squared terms we generate the formula library of Narendra et al. (2025, Sect. 4.2.3). The second-order (O2) library enumerates every non-empty subset of the 14 linear-plus-squared terms, wrapped in a pairwise-interaction operator, giving $2^{14} - 1 = \mathbf{16,383}$ formulae; the linear-only (O1) library gives $2^7 - 1 = \mathbf{127}$; a further **148** smoothed-term (GAM) formulae are generated separately. The total of

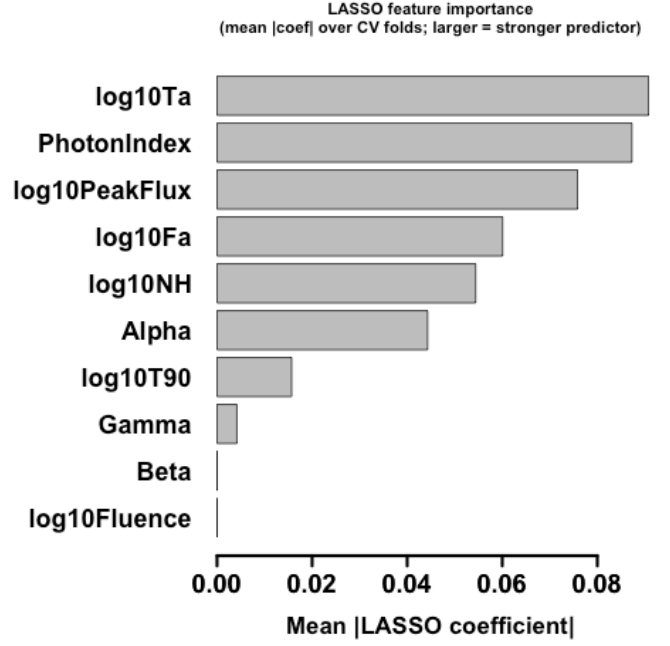


Fig. 2. Mean absolute LASSO coefficient of each candidate feature over 100 resamples. The seven highest-ranked features (above) are retained for formula generation and model training.

$127 + 16,383 = \mathbf{16,510}$ matches the count reported by Narendra et al. (2025). The search operates on the O2 library.

Each formula is evaluated over **100** randomised 80:20 train/test splits. Within a split, every formula receives a ten-fold cross-validation (10fCV) pass on the training set, and its correlation r and RMSE are computed in *linear* redshift space (predictions are transformed $z = 10^{\text{pred}} - 1$ before scoring), so that the selection metrics correspond to the linear axes of Fig. 6 of Narendra et al. (2025) and their cutoffs ($r = 0.565$, $\text{RMSE} = 1.167$). Formulae with r above the 99.9th percentile *and* RMSE below the 2nd percentile of the split are carried to the held-out test set. On the test set we additionally compute the bias $\langle z_{\text{pred}} - z_{\text{obs}} \rangle$ and use it as a *gate*: the worse half by $|\text{bias}|$ is discarded, after which the best- r and best-RMSE survivors each cast one vote. We adopt bias as a gate rather than an equal-weight vote because the single-split best- $|\text{bias}|$ formula scatters across nearly all 100 splits, indicating that the mean signed residual of one split is too noisy to vote on directly. Across the 100 splits the most-voted formula is

$$\begin{aligned} \log_{10}(z + 1) \sim & (\log F_a^2 + \log F_a + \log \text{PeakFlux})^2 \\ & + \log N_H + \text{PhotonIndex} + \log T_a + \alpha + \log T_{90} \\ & + \log \text{PeakFlux}^2 + \log N_H^2 + \text{PhotonIndex}^2 \\ & + \log T_a^2 + \alpha^2 + \log T_{90}^2. \end{aligned} \quad (3)$$

The search is checkpointed after every split and is fully resumable. Figure 3 reproduces the linear-scale RMSE- r plane over the full library with the selected candidates highlighted.

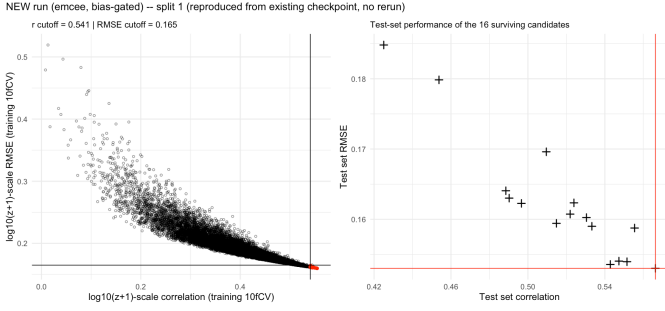


Fig. 3. Linear-scale RMSE versus correlation for the $\sim 16,500$ candidate formulae (analogue of Fig. 6 of Narendra et al. 2025). Highlighted points pass the $r \geq 99.9$ th-percentile and $\text{RMSE} \leq 2$ nd-percentile cutoffs and are carried to the held-out test set.

1.6. Robust removal of catastrophic outliers

Before ensemble training we apply the M-estimator cut of Narendra et al. (2025, Sect. 4.3). A Huber robust regression (`MASS::rlm`, method M) is fitted with the winning formula (Eq. 3), and the bursts in the lowest 5% of robustness weight are removed.

1.7. The SuperLearner ensemble

We train the stacked SuperLearner ensemble of Narendra et al. (2025, Sect. 4.4) in two configurations. The *baseline* configuration uses a broad learner library (elastic net, gradient-boosted trees, CART, MARS, bagged trees, neural network, random forests, stepwise regressions, and the tuned GAM/GLM/Bayesian-GLM learners built from Eq. 3), imputes missing values with multiple imputation by chained equations (MICE; van Buuren & Groothuis-Oudshoorn 2011) using the predictive-mean-matching “midastouch” method with $m = 20$ (predictors and errors imputed separately), stacks by non-negative least squares, and is cross-validated with 10fCV repeated 100 times.

The *augmented* configuration adds several components aimed at exploiting the optical/X-ray heterogeneity. Missing predictors are co-imputed with `missForest` (Stekhoven & Bühlmann 2012) jointly with an unlabelled generalisation pool (which never sees the redshift), repeated over three imputation seeds and four projection draws and averaged. Domain membership is encoded through a Daumé-style augmentation (Daumé 2007): an `is_optical` indicator plus scaled per-domain copies of the predictors ($c = 0.25$). For optical-only bursts the directly measured prompt features are restored at run time from the optical catalogue (“real-value recovery”). The learner library additionally includes a band-interaction GLM (with `is_optical` interacting with the plateau terms) and a boosted additive model (`mboost::gamboost`), and stacking is performed with a non-negative ridge variant of the NNLS meta-learner. A leakage-safe isolation feature (the distance to each burst’s tenth nearest neighbour in standardised predictor space, fitted on training folds only) supplies an outlier signal to the ensemble.

Table 1. Cross-validated redshift-estimation metrics for the emcee-combined sample. “All” is the full cross-validated sample; “inside cone” is the subset whose true redshift falls within its 1σ Monte Carlo interval. All values provisional ($\dagger$, see text).

Configuration	Subset	N	$r_{\log}$	$\text{RMSE}_{\log}$	r_z
Baseline	all	216 $\dagger$	0.646 $\dagger$	0.140 $\dagger$	0.616 $\dagger$
Baseline	inside cone	58 $\dagger$	0.911 $\dagger$	0.057 $\dagger$	0.924 $\dagger$
Augmented	all	192 $\dagger$	0.641 $\dagger$	0.132 $\dagger$	0.640 $\dagger$
Augmented	inside cone	47 $\dagger$	0.886 $\dagger$	0.062 $\dagger$	0.891 $\dagger$

1.8. Monte Carlo error propagation

We propagate the input measurement uncertainties through the trained ensemble as in Narendra et al. (2025, Sect. 4.7). Each mapped feature is perturbed by drawing from $\mathcal{N}(\text{value}, \sigma_{\text{catalog}})$ with σ_{catalog} treated as a 1σ standard deviation, the corresponding squared terms are regenerated, and 10^3 realisations are pushed through the ensemble. For each burst we report the median prediction, the 68% (“ 1σ ”) interval, and whether the true $\log_{10}(z + 1)$ lies inside that interval (“inside the cone”). Throughout, the metrics follow the parent paper: the correlation r , RMSE, bias, and the normalised median absolute deviation $\text{NMAD} = \text{median} |\Delta z / (1 + z_{\text{obs}})|$, together with the $1\sigma/2\sigma$ cone half-widths defined from the standard deviation of the log-scale residual $\Delta \log_{10}(z + 1)$.

2. Results

All metrics in this section are marked provisional ($\dagger$): they derive from the completed formula search (Eq. 3) and the current ensemble runs, while a linear- z re-derivation of the winning formula is still in progress and may shift the selection.

2.1. Cross-validated redshift estimation

Table 1 summarises the cross-validated performance of the two SuperLearner configurations on the emcee-combined sample. The baseline ensemble reaches $r = 0.646\dagger$ in $\log_{10}(z + 1)$ ($\text{RMSE} = 0.140\dagger$) over $N = 216$ bursts, coinciding with the headline correlation of Narendra et al. (2025) despite the enlarged and more heterogeneous training sample. The augmented configuration attains a comparable $r = 0.641\dagger$ at a lower log-scale RMSE ($0.132\dagger$) and a markedly lower linear- z RMSE ($0.975\dagger$ versus $1.070\dagger$), after removing eight catastrophic 2σ outliers (and, upstream, 19 bursts with physically impossible imputed feature values). Figures 4 and 5 show the predicted-versus-observed relations.

For reference, retraining the baseline ensemble on the previous (pre-search) formula yields only $r = 0.548\dagger$, so the formula search of Sect. 1.5 accounts for a substantial fraction of the performance.

2.2. Formula-search outcome

Over the 100 splits the search cast 300 votes across 118 distinct formulae. The winning formula (Eq. 3) received 28 votes, well clear of the runners-up, and is the formula used to train both ensemble configurations above. Figure 6 shows

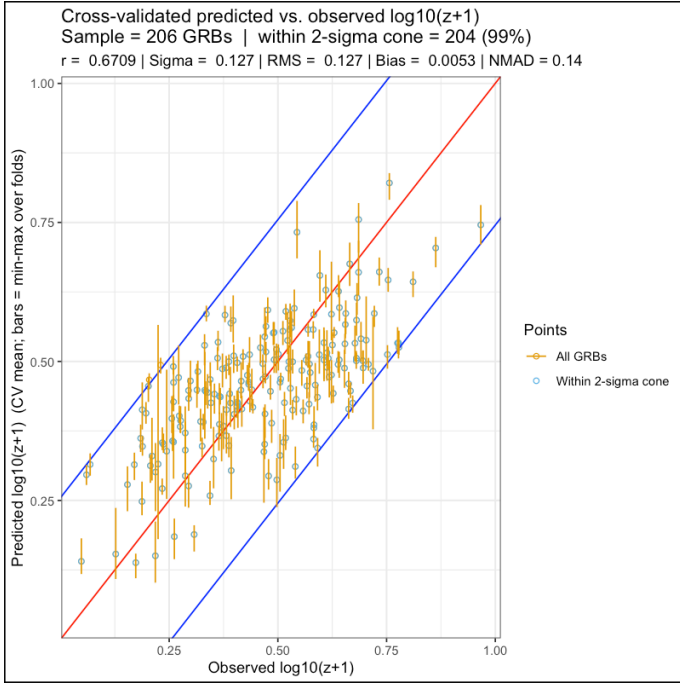


Fig. 4. Predicted versus observed $\log_{10}(z+1)$ for the baseline SuperLearner configuration (catastrophic outliers removed).

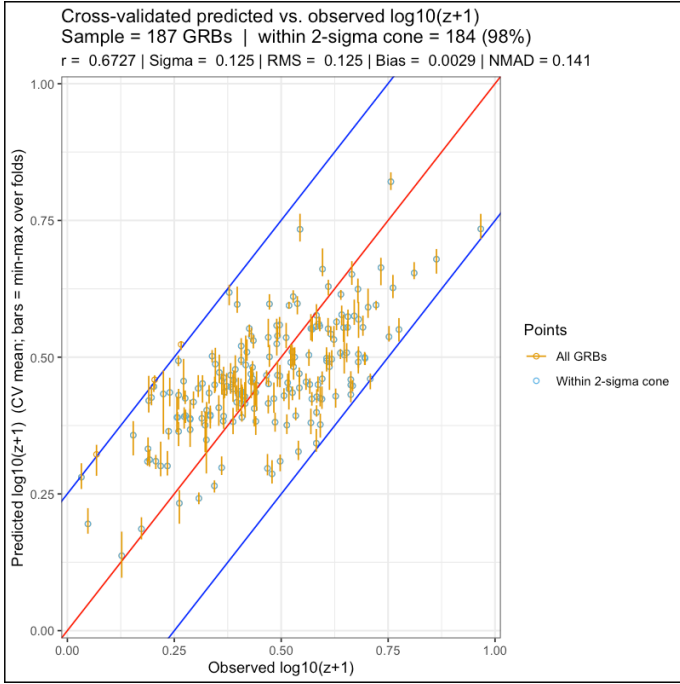


Fig. 5. Predicted versus observed $\log_{10}(z+1)$ for the augmented SuperLearner configuration (catastrophic outliers removed).

the surviving candidates in the RMSE- r plane sized by vote count.

2.3. Monte Carlo calibration

Within the subset of bursts whose true redshift falls inside its 1σ Monte Carlo interval, the correlation rises sharply — to $r = 0.911^\dagger$ (baseline, $N = 58$) and $r = 0.886^\dagger$ (augmented, $N = 47$) — demonstrating that where the in-

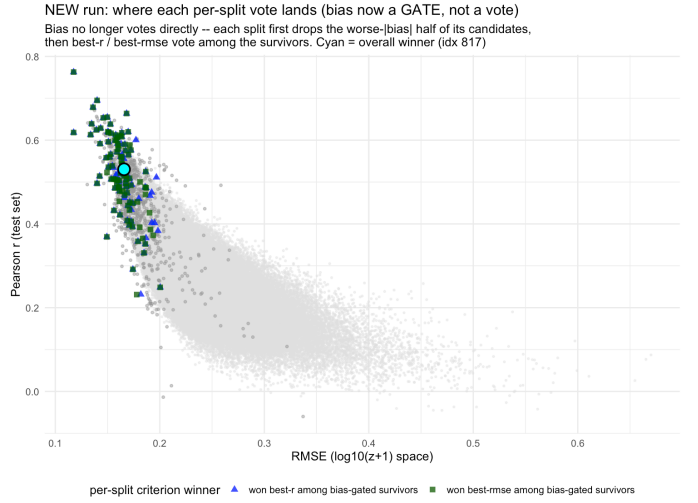


Fig. 6. Surviving candidate formulae in the linear-scale RMSE- r correlation plane, with marker size proportional to the number of cross-split votes received.

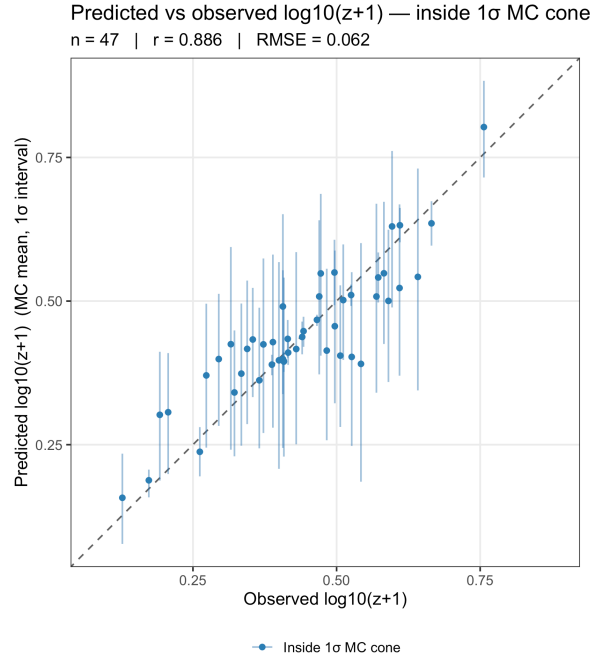


Fig. 7. Monte Carlo predicted versus observed $\log_{10}(z+1)$ for the augmented configuration, restricted to bursts inside their 1σ interval, with per-burst 1σ error bars.

put measurement errors are well controlled the ensemble is strongly predictive (Fig. 7). The nominal interval coverage is currently low (26.9%[†]/24.5%[†] of bursts inside the nominal 95% interval), indicating that the propagated intervals are under-dispersed; recalibrating the Monte Carlo intervals is one item being addressed in the ongoing rerun.

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